The Fold:
Scale, twist, collapse — and what is left at the centre

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0 Summary

Scaling about a datum is a one-parameter family $g_k(P) = P_0 + k(P - P_0)$. The family is a group for $k \neq 0$; $k = 0$ is the boundary where it degenerates into a point. A real k changes size, an imaginary one twists, and a fold is the reflection whose composition produces the twist. Everything that follows is the consequence of one determinant:

$$\det(g_k) = k^n, \qquad n = \text{dimension}. \tag{1}$$

Heights scale like k, areas like k^2, volumes like k^3. That single fact separates every quantity in the construction.

1 The scale factor and the datum

For a datum point P_0 and a point P on the object:

$$P' = P_0 + k(P - P_0). \tag{2}$$

k	behaviour
$k > 1$	expansion away from the datum
$0 < k < 1$	contraction toward the datum
$k = 0$	collapse: every point maps to P_0
$k < 0$	scaling composed with reflection through P_0

The map is a **homothety**. For $k \neq 0$ the family is a group isomorphic to $\mathbb{R}^*$; g_0 is the constant map with rank 0 and is not a member. This distinction matters later: zero is not a small scaling, it is the end of the family.

Area and height

$$\text{height} \quad h' = k\,h, \tag{3}$$

$$\text{area} \quad A' = k^2\,A, \tag{4}$$

$$\text{volume} \quad V' = |k|^3\,V, \tag{5}$$

so that

$$\frac{A'}{A} = \left(\frac{h'}{h}\right)^2. \tag{6}$$

Height and area cannot share a law. A sign survives in the height and dies in the area, because area is a determinant and the determinant of kI is k^2. At $k = -1$ the height reverses and the area is unchanged.

2 Two regimes: the sphere and the cone

Where the datum sits decides the geometry, and there are exactly two cases.

Datum inside the body. Every orbit of the scaling is a diameter. The level sets are spheres:

$$g_k(S_r) = S_{|k|r} \qquad (k \neq 0). \tag{7}$$

The tangent cone at an interior point is the whole space $\mathbb{R}^3$ — no proper cone exists. The flow is purely radial and the collapse is isotropic. The bulk-to-boundary ratio vanishes:

$$\frac{V}{A} = \frac{r}{3} \to 0. \tag{8}$$

Datum on the boundary. The orbits are rays and their union is a proper cone with apex P_0. This is the case that produces the cone of rays, and the case a horizon belongs to.

Datum outside. Same cone, seen from further away: the body subtends a cone and casts a shadow.

The three named choices of datum are three positions of the same apex:

datum	apex	tangent cone	behaviour
centroid	interior	$\mathbb{R}^3$	uniform scaling in place
vertex / base	boundary	proper cone	apex locked, body grows away
origin	fixed frame point, interior or exterior	$\mathbb{R}^3$ or proper cone	scaling also translates the body when it is off-origin

3 The twist

The operator

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I \tag{9}$$

is the infinitesimal generator of rotation, and

$$e^{\theta J} = \cos\theta\, I + \sin\theta\, J = R(\theta). \tag{10}$$

A general scale-with-twist is therefore the spiral similarity

$$k = \rho e^{i\theta}, \qquad P' = P_0 + \rho\,(\cos\theta\, I + \sin\theta\, J)\,(P - P_0). \tag{11}$$

J itself is the quarter turn. The four cardinal values are the fourth roots of unity and form a cycle:

$$1 \to i \to -1 \to -i \to 1. \tag{12}$$

Two quarter turns equal one half turn: $i^2 = -1$. The twist squared is the flip.

Which twists are observable

The twists that leave a body unchanged *are* its rotation symmetry group. Written as a quotient, the body only responds to

$$U(1)/\operatorname{Sym}(B). \tag{13}$$

body	symmetry	observable twist
sphere	S^1 — all θ	none: $i = 1$ exactly
cube	$C_4 = \{1, i, -1, -i\}$	only between the cardinal turns
chiral body	$\{1\}$	every angle

Measured on the rendered model (silhouette change under θ):

θ	0°	30°	45°	90°	180°	270°
sphere	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
cube	0.0%	1.4%	**12.3%**	0.0%	0.0%	0.0%

The peak at 45° is the antipode of the fundamental domain: it is the point furthest from any symmetry.

Orientation

Because $\det(kI) = k^n$, the sign of k has different meaning in different dimensions:

- **2D**: $\det(-I) = +1$ — $k = -1$ is a rotation by 180°, orientation preserved.
- **3D**: $\det(-I) = -1$ — $k = -1$ is a point reflection, orientation reversed: the genuine "inside out".

The same formula, two different geometries.

4 The fold, completed

A **fold** is a reflection in a hyperplane H:

$$\sigma_H^2 = I, \qquad \sigma_H \text{ fixes } H, \qquad \det \sigma_H = -1. \tag{14}$$

Its fixed set has dimension $n - 1$: a line in the plane, a plane in space. This is the piece the earlier account was missing — the fold does not fix the datum, it fixes a **crease**.

Two folds make a twist. Composing reflections in two hyperplanes meeting at angle α gives a rotation by 2α about their intersection:

$$\sigma_{H_2} \circ \sigma_{H_1} = R(2\alpha). \tag{15}$$

So:

- J (90°) is **two folds at** 45° — never a single fold
- $-I$ in the plane (180°) is two folds at 90°, or equivalently J^2
- J is orientation-preserving because it is the product of an **even** number of folds

Orientation is the parity of the fold count. This is why a twist and a fold cannot be confused: one has determinant $+1$, the other -1, and any odd number of folds reverses orientation.

The hinge is the intersection. The twist fixes the intersection of the two creases: a point in the plane, a line in space. The datum is that hinge.

The degeneration. Send the two creases together and their intersection collapses to a point. The fold then has no crease left, and what remains is $k = -1$: a point reflection. That

is the precise content of "the datum becomes the hinge of the fold" — the crease has shrunk into the datum, and the fold has become the point reflection.

The crease is the midpoint locus. Halve the segment from a point to its image. If $g_k(P) = P_0 + k(P - P_0)$ then

$$m_k(P) = \tfrac{1}{2}\big(P + g_k P\big) = g_{(1+k)/2}(P), \qquad \det m_k = \left(\frac{1+k}{2}\right)^n. \tag{16}$$

The midpoint map is itself a homothety about the same datum, and its determinant vanishes at $k = -1$ and nowhere else, in every dimension. So of the whole family exactly one member puts every midpoint on the datum, and that member is the point reflection. For an honest fold σ_H the midpoint map is $P \mapsto \frac{1}{2}(P + \sigma_H P) = \pi_H$, the orthogonal projection onto the crease: rank $n-1$, not rank 0. The crease is therefore the fixed set of the midpoint map, and the datum is the crease of zero dimension. Two other members are worth naming: $k = \frac{1}{2}$ makes the image the midpoint of datum and point, and $k = 2$ makes the point the midpoint of datum and image — the reciprocal pair, in which the halving is applied at either end of the same segment.

Any division point obeys the same law. At fraction t of the way from P to $g_k P$,

$$q_t(P) = (1 - t + tk)\, P, \tag{17}$$

again a homothety, now collapsing at $k = (t-1)/t$. Every $k \neq 1$ is therefore owned by exactly one division point, $t = 1/(1-k)$, and $t \leftrightarrow 1-t$ pairs k with $1/k$; the midpoint is the self-paired case. The quarter point owns $k = -3$, and there $m_{-3} = g_{-1}$ — the fold itself.

Completion. The fold group closes when the creases have swept the full angle:

$$(\sigma_2\sigma_1)^4 = I \qquad \text{for } \alpha = 45^\circ. \tag{18}$$

Four quarter turns return the body to itself; the two creases at 45° produce the four reflection axes at multiples of 45°, and the dihedral group D_4 of order eight is generated by exactly two folds. The cycle $1 \to i \to -1 \to -i \to 1$ is that closure written multiplicatively.

The only set fixed by the entire group is the hinge: a point in the plane, a line in space.

5 The horizon

A horizon is the boundary case of section 2: the apex sits on the surface, and the surface becomes null. It is where the cone reappears.

Under $r \to kr$ the mass scales too, because $r_s = 2GM/c^2$ is a length. That family is a set of solutions, not a process: every member has the same compactness, so the horizon is present in all of them or in none. A collapse is the opposite — the mass is held fixed and only R falls. The two give different laws and must not be read off one table:

quantity	homothetic family, $M \propto k$	collapse, M fixed
horizon radius r_H	$\propto k$	$= r_s$, constant
horizon area A_H	$\propto k^2$	constant
entropy $S = A_H/4\ell_P^2$	$\propto k^2$	constant
temperature T	$\propto 1/k$	constant
mass M	$\propto k$	constant
body radius R	$\propto k$	$\propto k$
compactness r_H/R	constant	$\propto 1/k$

Only the right-hand column is a process. In the left-hand column r_H/R is a label of the member, not a stage, and the first law is exact: with $S = \alpha M^2$ and $T = \beta/M$, $T\,dS = dM$ holds identically, and $S \cdot T \propto k$ is the horizon's energy. In the right-hand column M is fixed, so

both sides of $T\,dS = dM$ vanish once the horizon exists: the entropy is not accumulated during the fall, it appears with the horizon in one step.

The horizon arrives **before** zero. A body of radius R reaches its own Schwarzschild radius at

$$k^* = \frac{r_s}{R} = \frac{2GM}{c^2 R}. \tag{19}$$

body	k^*
Earth	1.4×10^{-9}
Sun	4.3×10^{-6}
70 kg water sphere	4×10^{-25}

Below k^* the collapse has left the observable chart. The point remains in the mathematics and leaves the physics.

Because $V \propto k^3$ while the horizon area is constant, $V/A_H \propto k^3$ in the collapse, and at the moment of formation it is $r_s/3$; in the homothetic family both scale, so the ratio falls only as k. Either way $V/A \to 0$. That is the quantitative content of the holographic statement: information capacity tracks the area, and the area outlives the volume.

6 Zero as the point

$k = 0$ is not a scaling. It is:

- **outside the group** — rank 0, not invertible
- **the compactification of every orbit** — each ray $(P_0, P]$ is open, and $k = 0$ closes them all at once
- **without geometry** — no tangent space, no normal; curvature $1/r$ is undefined, not infinite

The dimensional ladder is therefore not a path through intermediate objects:

$$V \propto k^3, \qquad A \propto k^2, \qquad h \propto k, \qquad \text{point} = k^0. \tag{20}$$

These are four measurements of the same family, not four consecutive states. At $k = 0$ the rank drops from n to 0 in one step; the body never passes through a sheet or a line.

Seen through invariants the same statement has a sharper form. Scaling keeps direction and the cross-ratio and destroys length; folding keeps length and angle and destroys direction; the two groups meet only in $\{I, -I\}$, and $k = -1$ is the one scaling that is also an isometry. Allow both and the action on $\mathbb{R}^n \setminus \{P_0\}$ is transitive: the invariant ring is the constants. That is the precise sense in which the point is what is left when everything is allowed.

7 What an observer can see

A point has no aperture: it subtends zero solid angle and collects zero flux. What it *can* measure is fixed by its dimension:

observer dimension	observable
0D	scalar — curvature, potential, temperature
1D	derivative, direction
2D	flux (needs an oriented area)
3D	density, integral

A point sees **values, not flows**. It has no tangent space, so it cannot host a vector. Two complementary views of the same sphere:

observer	$\Omega/4\pi$	behaviour
outside	5.06% at $k = 1$, falls as k^2	body shrinks to a point
at the centre	100% for every $k > 0$	sky fixed, distance shrinks
on the surface	2π half-space	edge-on; the horizon case

Measured exterior view:

k	$\Omega/4\pi$	ratio
1.00	5.060%	1.000×
0.50	1.216%	0.240×
0.25	0.301%	0.060×
0	0.000%	—

The interior view is constant because a homothety is **conformal**: it preserves angles, hence solid angle, exactly. Scaling preserves what you see, but not how far away it is. Let the interior sphere recede ($k \to \infty$) and the curvature $1/r \to 0$ while $\Omega = 4\pi$ stays fixed — the boundary flattens into the celestial plane at infinity, still carrying everything the bulk can send.

That is the sphere's case, where the rim is at 90° in every direction. A body whose range is a cone — apex at the centre, rim at the range, aperture θ — covers only $\Omega/4\pi = \sin^2(\theta/2)$, and the fraction is again independent of k: the same conformity, now with a rim. The sphere is the member with $\theta = 180°$; it is also the only member the fold fixes, since $k = -1$ carries every proper cone to its antipode — same aperture, opposite sky.

At $k = 0$ all three views coincide: the point, the shell, and the observer are the same place.

8 Devices that instantiate each step

idea	device	what it demonstrates
scale invariance	pinhole camera	image scales as k, area as k^2
interior collapse	inertial confinement, sonoluminescence	compression $1/k^3$
radial flow	integrating sphere	$\Omega = 4\pi$ independent of radius
twist J	quarter-wave plate	Jones matrix $\mathrm{diag}(1, i)$
two folds make a twist	two mirrors at 45°	rotation by twice the mirror angle
symmetry quotient	square waveguide	invisible at multiples of 90°
symmetry breaking	chiral medium (sugar solution)	every angle observable
directional twist	Faraday rotator	non-reciprocal 90°
horizon	acoustic / optical analogue horizon	$r \propto k$, $T \propto 1/k$
conformal boundary	AdS/CFT, celestial sphere	physics at infinity, Ω fixed
0D scalar probe	thermocouple, curvature scalar	value without direction

Companion models: nine HTML files in the repository, one per step of the paper.

9 The one sentence

Scaling preserves angles and curvature-free structure, changes lengths and curvatures, is generated by folds in pairs, and terminates in a point that has no geometry and can only be seen as a direction.